Closed loop control of three-phase AC-DC converter-IV

Introduction

In the rapidly evolving landscape of charging infrastructure, efficient and reliable power conversion is paramount. Three-phase AC-DC converters, often found at the heart of electric vehicle charging stations or industrial power supplies, demand sophisticated control mechanisms to ensure stable operation and optimal performance. This lecture delves into the intricate world of closed-loop control for these converters, building upon foundational concepts to achieve precise output voltage regulation and maintain a unity power factor when drawing power from the grid.

The journey through closed-loop control begins with understanding how to manage complex, interdependent three-phase currents and voltages. By transforming these quantities into more manageable, independent components within specialized reference frames, we unlock the ability to design robust controllers. This lecture systematically breaks down the design of both inner current loops and outer voltage loops, detailing the mathematical derivations and practical considerations for setting controller parameters. Ultimately, mastering these control strategies is crucial for developing high-performance, grid-friendly power electronics essential for the future of energy.

Core Concepts

Term	Definition	Significance
`DQ Reference Frame`	A synchronously rotating coordinate system (d-axis and q-axis) used to transform three-phase AC quantities into two DC quantities, simplifying control.	Enables independent control of active and reactive power components, crucial for regulating output voltage and power factor.
`Unity Power Factor`	A condition where the current drawn from the grid is perfectly in phase with the voltage, minimizing reactive power and maximizing efficiency.	Essential for grid compatibility, reducing losses, and ensuring efficient energy transfer in AC-DC conversion.

Term	Definition	Significance
`PI Controller`	A Proportional-Integral controller, a common feedback control loop mechanism that calculates an error value as the difference between a measured process variable and a desired setpoint.	Provides both fast response (proportional) and elimination of steady-state error (integral), widely used for regulating voltage and current in power converters.
`Phase-Locked Loop (PLL)`	An electronic circuit that generates an output signal whose phase is related to the phase of an input signal, used here to synchronize with the grid voltage.	Crucial for aligning the DQ reference frame with the grid voltage vector, enabling accurate control of active and reactive power.
`Cross-Coupling Terms`	Interdependencies between the d-axis and q-axis components in the DQ model of an AC system, often appearing as ωLIq or ωLId.	If not compensated, these terms introduce undesirable interactions between control axes, complicating independent control and degrading performance.
`Voltage Space Vector`	A mathematical representation that combines the three-phase voltages into a single rotating vector in a complex plane.	Provides a unified view of the three-phase voltage system, simplifying analysis and control in rotating reference frames.
`Current Space Vector`	Similar to the voltage space vector, this represents the three-phase currents as a single rotating vector.	Its alignment with the voltage space vector is key to achieving unity power factor operation.
`Bandwidth`	The range of frequencies over which a system or component operates effectively, often characterized by a pole in its transfer function.	Determines the speed of response and dynamic performance of sensors and control loops; higher bandwidth generally means faster response.

Term	Definition	Significance
`Phase Margin`	A measure of stability in a feedback control system, indicating how much the phase of the open-loop transfer function can change before instability occurs.	A larger phase margin (e.g., 30-60 degrees) ensures robust and stable closed-loop operation, preventing oscillations.

Detailed Analysis

1. Control Objectives and Reference Frame Transformation

The primary objectives for the three-phase AC-DC converter are twofold: first, to regulate the output DC voltage to a desired value, and second, to maintain unity power factor operation when drawing current from the grid. Achieving these objectives in a three-phase system is complex because the phase currents and voltages are interdependent. To overcome this, a transformation into a more manageable reference frame is necessary.

Initially, the stationary alpha-beta reference frame is introduced, where the voltage space vector rotates. The alpha and beta components are 90 degrees apart, allowing for independent control. However, for even simpler control, the synchronously rotating DQ reference frame is preferred. In this frame, the d-axis is aligned with the rotating voltage space vector, effectively making the q-axis component of the voltage (Vsq) zero. This alignment simplifies the control problem by converting AC quantities into DC quantities, which are easier to regulate. Similarly, by ensuring the q-axis component of the current (ISQ) is zero, the current space vector aligns with the d-axis, which is already aligned with the voltage space vector, thereby achieving unity power factor.

2. DQ Axis Model and Cross-Coupling Compensation

The mathematical model of the converter in the DQ reference frame reveals cross-coupling terms (e.g., ωLISQ and ωLISD) that link the d-axis and q-axis dynamics. These terms, if unaddressed, would prevent independent control of the d and q components. To achieve true independent control, feed-forward compensation is employed. These feed-forward terms are added to the controller outputs to cancel out the cross-coupling effects, effectively decoupling the d-axis and q-axis control loops. This results in two independent control loops: one for the d-axis current (ISD) and one for the q-axis current (ISQ).

3. Current Loop Design and Parameter Calculation

The inner current loops (for ISD and ISQ) are designed using PI controllers. A standard PI controller has a transfer function of KPC + KIC/S. For the current loop, a common design approach involves pole-zero cancellation. The plant (the L-Rs series impedance) has a pole at -Rs/L. By setting the zero of the PI controller (KIC/KPC) equal to Rs/L, this pole can be cancelled, simplifying the system dynamics.

The overall closed-loop transfer function for the current loop, considering the converter, current sensor, and PI controller, is then analyzed. Small time constants associated with switching delay (TD) and current sensor bandwidth (T1) are approximated. For instance, the product of TD and T1 is considered negligible, simplifying the denominator terms. The resulting transfer function is then compared to a standard second-order system (S^2 + 2ζωnS + ωn^2) to determine the proportional gain (KPC) and integral gain (KIC). By choosing a damping factor (ζ) of 0.707 (for optimal transient response), the proportional gain KPC can be derived as:
KPC = L / (2 * G * K1 * Tsigma)
where G is the converter gain, K1 is the current sensor gain, and Tsigma is the sum of TD and T1. The integral gain KIC is then found from the pole-zero cancellation condition: KIC = KPC * (Rs/L).

4. Voltage Loop Design and Parameter Calculation

The outer voltage loop is designed to be significantly slower than the inner current loop. This hierarchical control ensures that the inner current loop can quickly track any changes commanded by the outer voltage loop. Due to the slower nature of the voltage loop, the dynamics of the inner current loop can be approximated as a simpler transfer function (e.g., 1/K1) for the purpose of voltage controller design.

The voltage controller also uses a PI structure (KPV + KIV/S). The open-loop transfer function of the voltage control system is then analyzed using Bode plots. To ensure stability and a good phase margin, the gain plot is designed to cross the 0 dB line with a slope of -20 dB per decade. This is achieved by strategically placing the zero of the voltage controller (KIV/KPV) and the poles introduced by the system (e.g., capacitor dynamics, sensor bandwidth). By setting the zero of the voltage controller at a frequency lower than the dominant pole, the desired slope can be achieved. A factor 'a' is introduced, where KPV/KIV = a^2 * Tdelta (Tdelta being a combined time constant from the current loop and voltage sensor). A common choice for 'a' is 2, which typically yields a phase margin of approximately 37 degrees, ensuring stable operation. The proportional gain KPV is then derived based on the desired crossover frequency and system parameters:
KPV = c * K1 / (K2 * Ka * Tdelta)

where 'c' is a constant, K1 and K2 are sensor gains, and Ka is a gain factor. The integral gain KIV is then determined from the relationship with KPV.

5. Power Balance and Sensor Gain Determination

To complete the control system design, the 'K' factor, which relates the converter's output current to the d-axis current, and the sensor gains (K1, K2) must be determined. The 'K' factor is derived from the principle of power balance. For unity power factor operation, the three-phase power (3 * Vs_peak * Is_peak / 2) is equated to the two-phase power (VSD * ISD + VSQ * ISQ). Since VSQ and ISQ are ideally zero, the power simplifies to VSD * ISD. This leads to the relationship for the converter current (I_converter) as:
I_converter = (2/3) * (VSD / V0) * ISD
Thus, the 'K' factor is (2/3) * (VSD / V0).

Sensor gains (K1 for current, K2 for voltage) are determined by the ratio of the maximum voltage acceptable by the microcontroller (e.g., 3.3V, 1.8V) to the maximum expected value of the sensed quantity (e.g., 3/2 * max_supply_current for K1, max_output_voltage for K2). These gains scale the physical measurements to a range suitable for the digital controller.

6. Overall Closed-Loop Control System and Phase-Locked Loop (PLL)

The complete closed-loop control system integrates all these components. The output voltage (V0) is sensed, fed back, and compared to a reference. The voltage controller generates the ISD reference. The ISD and ISQ (set to zero for unity power factor) references are fed into their respective current controllers. These controllers, along with feed-forward terms for cross-coupling compensation, generate the converter voltage commands (V_converter_D, V_converter_Q). These DQ voltages are then transformed into three-phase modulating waves (Vma, Vmb, Vmc) using the DQ to ABC transformation. A modulator compares these modulating waves with a triangular carrier signal to generate the gate signals for the converter switches.

Crucially, the entire system relies on accurate synchronization with the grid. This is achieved by a Phase-Locked Loop (PLL). The PLL senses the three-phase grid voltages (Van, Vbn, Vcn), converts them to the DQ frame (VSD, VSQ). By driving the VSQ component to zero through a PI controller, the d-axis is forced to align with the grid voltage vector. The output of this controller, combined with a feed-forward term for the nominal grid frequency, represents the angular speed (ω) of the grid. Integrating this ω yields the phase angle (θ), which is essential for all DQ transformations (ABC to DQ and DQ to ABC). This θ ensures that the converter's operation is perfectly synchronized with the grid, enabling stable unity power factor control. This comprehensive approach is often referred to as vector control.

Key Takeaways

- [ ] Closed-loop control of three-phase AC-DC converters aims to regulate output voltage and achieve unity power factor.
- [ ] The DQ reference frame simplifies control by transforming AC quantities into DC, allowing independent control of active and reactive power.
- [ ] Cross-coupling terms in the DQ model must be compensated using feed-forward terms to decouple d-axis and q-axis control.
- [ ] Current controllers (PI) are designed using pole-zero cancellation to simplify dynamics and achieve desired gains (KPC, KIC).
- [ ] Voltage controllers (PI) are designed to be slower than current loops, using Bode plot analysis to ensure stability and a good phase margin (e.g., 37 degrees).
- [ ] Sensor gains (K1, K2) are crucial for scaling physical measurements to microcontroller-compatible voltage levels.
- [ ] The 'K' factor, derived from power balance, links the converter's output current to the d-axis current.
- [ ] A Phase-Locked Loop (PLL) is essential for synchronizing the DQ reference frame with the grid voltage, generating the necessary phase angle (θ) for transformations.
- [ ] The overall control strategy, integrating voltage and current loops with DQ transformations and PLL, is known as vector control.
- [ ] Approximations of small time constants (e.g., switching delay, sensor bandwidth) are used to simplify transfer functions during controller design.

Conclusion

The detailed exploration of closed-loop control for three-phase AC-DC converters reveals a sophisticated interplay of electrical engineering principles and control theory. From the initial transformation of complex three-phase signals into manageable DQ components to the meticulous design of PI controllers for both current and voltage loops, every step is critical for achieving high-performance power conversion. The integration of feed-forward compensation for cross-coupling terms and the indispensable role of the Phase-Locked Loop for grid synchronization underscore the precision required in modern power electronics.

This comprehensive control strategy, often termed vector control, is not merely an academic exercise; it forms the backbone of reliable and efficient power systems. In real-world applications like electric vehicle charging, renewable energy integration, and industrial power supplies, these converters must operate seamlessly with the

grid, drawing power with a unity power factor to minimize losses and ensure grid stability. By understanding and implementing these closed-loop control techniques, engineers can design robust and intelligent power converters that are fundamental to the advancement of sustainable and efficient energy infrastructure.